Bio-inspired Climbing Robot with Dual Locomotion Strategies and Vision-based Manipulator Control

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I. Problem and Background

Achieving reliable locomotion on inclined, low-friction, and structured surfaces remains a fundamental challenge in robotics. In many real-world scenarios, such as inspection, search and rescue, and extraterrestrial exploration, robots are required to traverse complex terrains where traditional wheeled or legged locomotion strategies become ineffective. In particular, surfaces with discrete geometric features but low friction—such as structured plastic materials or artificial ridged environments—pose a unique difficulty: while geometric engagement is possible, the lack of friction severely limits the ability to maintain stable contact.

To address this challenge, our project explores a bio-inspired approach based on the locomotion mechanism of caterpillars. Caterpillars move forward through a cyclic sequence of extension, anchoring, and contraction. Specifically, the front part of the body extends forward to establish contact with a surface, followed by the contraction of the rear body, effectively pulling the entire body forward. This motion relies not on continuous rolling or stepping, but on discrete anchoring and pulling actions, which makes it particularly suitable for environments where friction is limited but geometric contact points are available.

Inspired by this locomotion strategy, we aim to design a robotic system capable of climbing structured, low-friction surfaces by leveraging controlled contact and pulling mechanisms. In Checkpoint 1, we formally defined this problem as a climbing task on an inclined wall with regularly spaced ridge-like protrusions. These ridges provide potential anchoring points, but their effectiveness depends heavily on the geometric interaction between the robot's contact mechanism and the surface. Rather than relying on friction-based adhesion, our approach focuses on **geometric interlocking**, where the robot engages with the ridges in a directional and mechanically constrained manner.

Through a review of existing literature, we identified that most prior work in climbing robotics falls into two main categories: microspine-based systems, which depend on rough surface asperities, and adhesive systems inspired by gecko locomotion, which rely on specialized materials and directional adhesion. However, both approaches are not well suited for structured but low-friction surfaces, such as the LEGO-based environment used in our experiments. This observation revealed a gap in existing research, motivating our investigation into simple, passive claw mechanisms capable of achieving reliable engagement through geometry alone.

Based on these considerations, we proposed a hypothesis that the performance of a claw-based climbing system depends strongly on both environmental parameters (such as slope angle and ridge geometry) and design parameters (such as claw length, curvature, and locking angle). This hypothesis formed the foundation for the experimental design and validation conducted in Checkpoints 2 and 3.

Building upon this initial problem formulation and hypothesis, we further expanded our design exploration to include multiple locomotion strategies. In particular, we investigated two distinct approaches:

- A **hook-based locomotion system**, which relies on passive, directionally compliant claws to achieve geometric interlocking with the surface.
- A **robotic arm-based locomotion system**, which introduces active control and additional degrees of freedom to enhance maneuverability.

By combining these two approaches, this project explores the trade-offs between passive mechanical design and active control strategies.

Overall, this work aims to bridge the gap between bio-inspired locomotion theory and practical robotic implementation.

II. Hook-Based Locomotion System

A. Design Overview

The hook-based locomotion system serves as the primary implementation of our bio-inspired climbing strategy. This design is directly motivated by the problem formulation established in Checkpoint 1, where we aim to enable reliable climbing on structured, low-friction surfaces through geometric interlocking rather than friction-based adhesion.

As discussed in the problem formulation, the target environment consists of an inclined surface with regularly spaced ridge-like protrusions. These ridges provide discrete anchoring points, but due to the low friction of the material (e.g., LEGO Technic plastic), traditional adhesion-based or friction-based methods are ineffective.

a) Mechanical Structure: The robot adopts a compact rectangular morphology with approximate dimensions of $20 \times 20 \times 10$ cm.

- A set of **directionally compliant claws**
- A **passive support system** composed of wheels

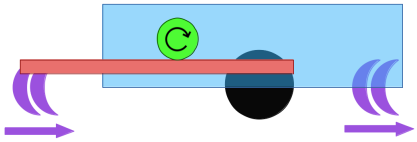


Fig. 1. Robot morphology and kinematic structure of the hook-based system.

b) Actuation Mechanism: The claw system is driven by a rack-and-pinion transmission.

1) Claws extend forward
2) Engage with ridges
3) Retract to pull body forward

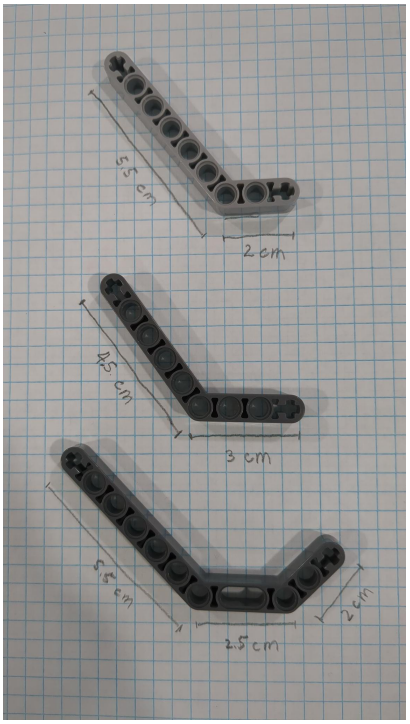


Fig. 2. Directional compliant claw mechanism

c) Design Rationale: The system is intentionally simplified to isolate claw geometry effects.

B. Research Question and Hypothesis

Building upon the problem formulation described in the previous section, the primary objective of this study is to investigate how a robot can achieve reliable locomotion on inclined, low-friction, and structured surfaces using passive mechanical interaction.

Specifically, we consider a climbing scenario in which the robot must traverse an angled surface populated with discrete ridge-like protrusions. These protrusions, with characteristic dimensions on the order of 1 cm and spacing ranging from approximately 2 cm to 10 cm, provide potential anchoring points for locomotion. The surface itself may be oriented at various incline angles, typically ranging from near-horizontal to moderately steep slopes. Under such conditions, friction-based locomotion becomes unreliable, and the robot must rely on geometric interaction with the environment to generate forward motion.

The central research question of this work can therefore be formulated as follows:

> *How can a robot utilize directionally compliant claw mechanisms to achieve stable and efficient locomotion on structured, low-friction inclined surfaces, and how do claw design parameters interact with environmental conditions to influence performance?*

To address this question, we adopt a design strategy based on **geometric interlocking**, in which the robot engages with surface features through passive mechanical constraints rather than relying on friction or adhesion. In this framework, the effectiveness of locomotion depends critically on the interaction between the claw geometry and the environmental structure.

Based on insights from prior work in bio-inspired climbing and mechanical engagement systems, we propose that both environmental parameters and claw design parameters play a dominant role in determining locomotion performance. The key environmental parameter considered in this study is the **slope angle** of the surface, while the primary design parameters include:

- Claw length
- Claw curvature
- Locking angle (engagement angle)
- Number and arrangement of claw contact points

Accordingly, we formulate the following hypothesis:

> *As the slope angle of the climbing surface increases, the optimal claw geometry must adapt to maintain stable geometric interlocking. Specifically, effective claw designs for steeper slopes are expected to exhibit shorter length, increased curvature, reduced locking angle, and a greater number of contact points to enhance stability and load distribution.*

This hypothesis is grounded in several key mechanical considerations. First, as the slope angle increases, the component of gravitational force acting parallel to the surface becomes larger, increasing the tendency of the robot to slip. To counteract this effect, the claw must form a more secure geometric constraint with the ridge, which typically requires a smaller effective engagement angle and a tighter locking configuration.

Second, increasing the curvature of the claw enhances its ability to wrap around the ridge structure, improving the robustness of the interlocking mechanism. Third, reducing the claw length helps prevent overextension and reduces sensitivity to positioning errors at higher inclines. Finally, increasing the number of contact points (e.g., through multiple claw rows or grid arrangements) improves load distribution and provides redundancy, reducing the likelihood of catastrophic failure due to local disengagement.

This hypothesis serves as the theoretical foundation for the experimental design presented in the following subsection.

In particular, it motivates a systematic investigation of how variations in slope angle and claw geometry affect locomotion performance metrics such as stability, slip frequency, and traversal success rate.

C. Experimental Setup

To validate the hypothesis proposed in the previous section, a series of controlled physical experiments were designed to systematically evaluate the locomotion performance of the hook-based robotic system under varying environmental and design conditions.

a) Experimental Environment: The experiments were conducted on a controlled inclined platform constructed using LEGO Technic components. The surface consists of regularly spaced ridge-like protrusions, which serve as discrete anchoring points for the claw mechanism. These ridges are designed to mimic structured, low-friction environments where geometric interlocking is possible but friction-based locomotion is ineffective.

The inclination of the surface can be adjusted to simulate different climbing conditions. In this study, three slope angles were selected:

- 7.5° (mild incline)
- 15° (moderate incline)
- 20° (steep incline)

These values were chosen to capture the transition from stable locomotion to failure-prone conditions.



Fig. 3. Experimental setup showing the inclined surface with ridge structures.

b) Robot Configuration: The robot used in the experiments follows the design described in the previous subsection. The system includes:

- A front-mounted directionally compliant claw mechanism
- A rack-and-pinion actuation system for cyclic extension and retraction
- Passive wheels for support and stabilization

During each experiment, the robot starts at the base of the inclined surface and attempts to traverse a fixed distance upward through repeated claw engagement cycles.

c) Experimental Variables: To evaluate the relationship between design parameters and locomotion performance, the following variables were considered:

Independent Variables:

- **Slope Angle (θ):** Controls the difficulty of the climbing task by increasing the gravitational load along the incline.
- **Wheel Assistance Mode:** Determines how auxiliary propulsion interacts with the claw mechanism. Three modes were tested:
  - *Wheels Off*: No active propulsion; locomotion relies entirely on claw engagement.
  - *Wheels Always On*: Continuous forward force is applied throughout the entire motion cycle.
  - *Wheels On Retract*: Wheels are activated only during the claw retraction phase.

Controlled Variables:

All other parameters were kept constant throughout the experiments to isolate the effects of the independent variables:

- Robot mass and geometry
- Claw material and basic structure
- Actuation speed and control timing
- Ridge geometry and spacing
- Initial position and orientation

d) Experimental Procedure: For each combination of slope angle and wheel mode, the robot performed multiple climbing trials. Each trial followed a standardized procedure:

1) The robot was placed at the base of the inclined surface.
2) The slope angle was set and verified.
3) The robot executed its climbing cycle using a predefined control strategy.
4) The trial ended when the robot reached the target height or experienced failure (e.g., slipping or falling).

Each condition was tested multiple times (approximately three repetitions per configuration) to ensure consistency and reduce experimental variability.

e) Performance Metrics: To quantitatively evaluate locomotion performance, the following metrics were recorded:

- **Traversal Success Rate**: Percentage of trials in which the robot successfully reached the target.
- **Slip Count**: Number of instances where the claw failed to maintain geometric interlock.
- **Completion Time**: Time required to complete successful climbs.
- **Remaining Distance**: Distance to the target at the end of unsuccessful trials.

These measurements enable a comprehensive assessment of both the efficiency and stability of the locomotion strategy, and provide the basis for validating the proposed hypothesis.

D. Results

A total of 26 trials were conducted to evaluate the locomotion performance of the hook-based robotic system under different slope angles and wheel assistance modes. The experimental conditions follow a 3×3 factorial design, consisting of three slope angles (7.5°, 15°, and 20°) and three wheel modes

(Wheels Off, Wheels Always On, and Wheels On Retract), with approximately three repetitions per condition.

a) Summary of Experimental Results: Table I summarizes the key performance metrics across all tested conditions, including traversal success rate (SR), mean slip count, and mean completion time for successful trials.

TABLE I
SUMMARY OF EXPERIMENTAL RESULTS ACROSS ALL CONDITIONS. SR = SUCCESS RATE (%), SLIP = MEAN SLIP COUNT, TIME = MEAN COMPLETION TIME (S) FOR SUCCESSFUL TRIALS ONLY.

Angle	Mode	SR (%)	Slip	Time (s)
7.5°	Off	67	3.7	291
7.5°	Always On	67	0.0	183
7.5°	On Retract	33	0.0	151
15°	Off	100	6.0	227
15°	Always On	33	2.0	197
15°	On Retract	33	2.0	121
20°	Off	33	9.3	334
20°	Always On	0	3.0	—
20°	On Retract	67	2.0	208

b) Slip Behavior: Figure 4 presents the measured slip count for all experimental conditions. A slip is defined as a failure of the claw to maintain geometric interlocking with a ridge, resulting in temporary loss of forward progress.

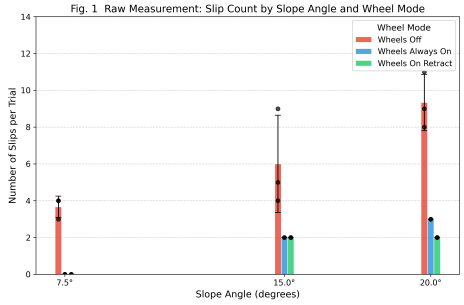


Fig. 4. Slip count per trial across slope angles and wheel modes. Error bars indicate ± 1 standard deviation.

Across all slope angles, the Wheels Off mode exhibits a significantly higher slip count compared to the two wheel-assisted modes. Furthermore, the slip count increases consistently with slope angle, indicating a strong dependence of stability on environmental difficulty.

c) Remaining Distance to Goal: Figure 5 shows the remaining distance to the target at the end of each trial. A value of 0 cm indicates successful completion.

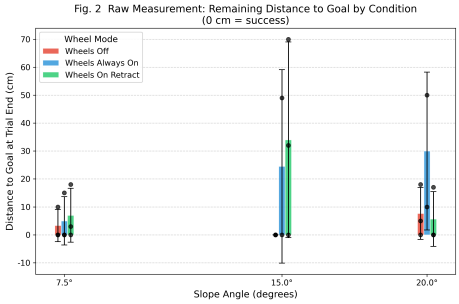


Fig. 5. Remaining distance to goal at trial termination. A value of 0 indicates successful traversal.

The Wheels Always On mode shows a large remaining distance at the highest slope angle (20°), indicating consistent failure under this condition. In contrast, the Wheels On Retract mode maintains lower remaining distances, even at steeper slopes.

d) Traversal Success Rate: Figure 6 illustrates the success rate as a function of slope angle for each wheel mode.

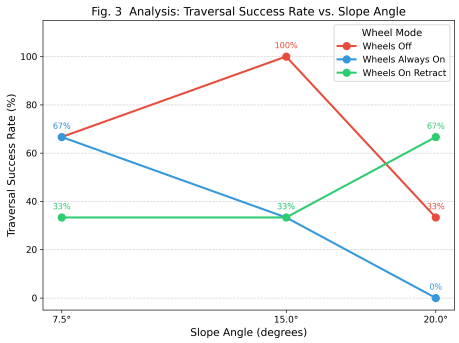


Fig. 6. Traversal success rate vs. slope angle for different wheel modes.

The results reveal a clear variation in performance across different configurations. The Wheels Off mode achieves its highest success rate at 15°, while the Wheels Always On mode shows a monotonic decrease in performance as slope angle increases. The Wheels On Retract mode demonstrates improved performance at steeper slopes.

e) Completion Time: Figure 7 compares the mean completion time for successful trials only.

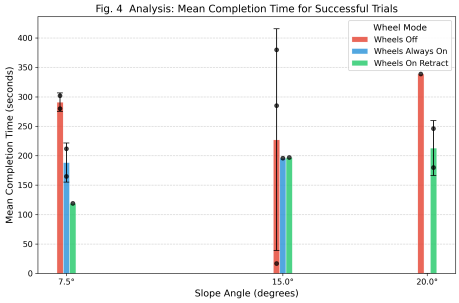


Fig. 7. Mean completion time for successful trials across different configurations.

The Wheels Off mode exhibits the longest completion times across all slope angles, reflecting the increased number of slips and recovery cycles. In contrast, the Wheels On Retract mode achieves consistently shorter completion times when successful, indicating higher locomotion efficiency.

f) Overall Observation: The collected data demonstrates a strong dependence of locomotion performance on both slope angle and wheel assistance mode. Significant differences are observed across configurations in terms of stability, efficiency, and success rate, highlighting the importance of both environmental conditions and system design in determining climbing performance.

E. Analysis

The experimental results presented in the previous section reveal several non-trivial trends that cannot be explained solely by intuitive assumptions about force generation. In this section, we analyze the underlying mechanisms that govern the observed performance differences across wheel modes and slope conditions.

a) Effect of Slope Angle: A clear trend observed across all configurations is that locomotion becomes increasingly difficult as the slope angle increases. This is reflected in the monotonic increase in slip count for the Wheels Off mode, where the average slip count rises from 3.7 at 7.5° to 9.3 at 20°.

This behavior can be explained by the increasing component of gravitational force acting parallel to the inclined surface. As the slope angle increases, the tangential force that must be resisted by the claw-ridge interaction becomes larger, placing higher demands on the geometric interlocking mechanism. Consequently, the system becomes more sensitive to small deviations in claw positioning and engagement, leading to a higher frequency of slips.

b) Role of Wheel Assistance: One of the most significant findings is the contrasting performance of different wheel assistance modes. While both wheel-assisted modes (Always On and On Retract) reduce slip count compared to the Wheels Off configuration, their impact on overall performance differs substantially.

In particular, the Wheels Always On mode shows a consistent degradation in performance as slope increases, ultimately resulting in complete failure at 20°. This result is counterintuitive, as continuous propulsion might be expected to assist climbing.

c) Failure Mechanism of Continuous Propulsion: The poor performance of the Wheels Always On mode can be explained by examining the kinematics of the claw engagement cycle. During the extension phase, the claws must move forward relative to the environment and establish contact with a new ridge. This phase requires a momentary reduction in forward body motion to allow proper alignment and engagement.

However, when continuous wheel propulsion is applied, the robot body is pushed forward even during the extension phase. This premature forward motion prevents the claws from properly engaging with the ridges, leading to failed interlocking and subsequent slipping. As a result, continuous propulsion actively disrupts the engagement mechanism, especially under high load conditions at steeper slopes.

d) Advantage of Phase-Synchronized Propulsion: In contrast, the Wheels On Retract mode demonstrates significantly improved performance at higher slope angles. This mode applies propulsion only during the retraction phase, when the claws are already securely engaged with the surface.

From a mechanical perspective, this phase-synchronized strategy aligns force application with the strongest point of the engagement cycle. During retraction, the claw-ridge interface forms a stable geometric constraint, allowing the robot to effectively convert actuation force into forward motion without risking disengagement.

This explains why the Wheels On Retract mode not only maintains low slip counts but also achieves higher success rates and shorter completion times at steeper slopes.

e) Key Insight: Timing vs. Force Magnitude: A critical insight emerging from this study is that the timing of auxiliary propulsion is more important than its magnitude. While it may seem intuitive that increasing propulsion force would improve climbing performance, the results demonstrate that improperly timed force application can be detrimental.

Specifically, continuous propulsion introduces destabilizing effects during the most sensitive phase of the locomotion cycle, whereas phase-aligned propulsion enhances performance by reinforcing existing mechanical constraints.

f) Transition in Optimal Strategy: Another important observation is the transition in optimal locomotion strategy as slope angle increases. At lower slopes (e.g., 7.5° and 15°), the Wheels Off mode performs comparably or even better in terms of success rate. However, at higher slopes (20°), the Wheels On Retract mode becomes the dominant strategy.

This suggests the existence of a crossover point beyond which passive geometric interlocking alone is insufficient, and carefully synchronized auxiliary propulsion becomes necessary for reliable climbing.

g) Summary of Findings: Overall, the analysis reveals that locomotion performance in structured, low-friction environments is governed by a complex interaction between mechanical design and control timing. The results highlight that:

- Increasing slope angle significantly amplifies instability and slip frequency
- Continuous propulsion can degrade performance by disrupting claw engagement
- Phase-synchronized propulsion enhances both stability and efficiency
- The optimal control strategy depends on environmental difficulty

These findings provide strong support for the hypothesis that effective locomotion in such environments depends not only on geometric design, but also on the coordination between actuation and contact dynamics.

III. Robotic Arm System

A. Design Motivation

While the hook-based locomotion system provides a simple and robust solution for climbing on structured, low-friction surfaces, its functionality is inherently limited by its passive nature. Specifically, the hook-based design relies on geometric interlocking along a single dominant direction, which restricts the robot's ability to generate forces outside of the predefined motion axis. As a result, the system lacks the capability to actively adjust its contact direction or explore alternative anchoring strategies when encountering complex or irregular terrain.

This limitation becomes particularly critical when the robot operates in environments where suitable anchoring points are not directly aligned with its forward direction. In such cases, the inability to apply lateral forces or reposition the contact mechanism significantly reduces the robot's maneuverability and adaptability, potentially leading to failure in maintaining stable locomotion.

To address these limitations, we introduce a robotic arm-based locomotion system as an alternative and complementary solution. Inspired by the same bio-inspired extension–anchor–pull principle, this design augments the system with active control and additional degrees of freedom, enabling the robot to interact with its environment in a more flexible and adaptive manner.

The key idea behind this design is to decouple the direction of force application from the primary body motion. By incorporating a multi-degree-of-freedom robotic arm, the end-effector (claw) can be positioned arbitrarily within a three-dimensional workspace. This allows the robot to:

- Reach and engage with off-axis anchoring points
- Apply forces in both forward and lateral directions
- Adapt to variations in terrain geometry

As a result, the robotic arm enables the system to actively select optimal contact locations and pulling directions, rather than relying solely on passive alignment with the environment.

Furthermore, the introduction of an actively controlled manipulator significantly enhances the controllability of the locomotion process. Instead of executing a fixed cyclic motion, the robot can dynamically adjust its movement strategy based on environmental feedback or high-level control inputs. This opens the possibility for more advanced behaviors, such as trajectory planning, interactive control, and adaptive gait generation.

From a system-level perspective, the robotic arm-based design represents a shift from a purely mechanical solution to a hybrid approach that integrates kinematic modeling, control algorithms, and real-time communication. While this increases system complexity, it provides a significantly higher degree of flexibility and generality.

Therefore, the robotic arm system is considered as a backup and extension of the primary hook-based solution. It serves both as a means to overcome the limitations of passive locomotion and as a platform for exploring more advanced control strategies in robotic mobility.

B. Planar Kinematics

To enable precise control of the robotic arm in three-dimensional space, we first establish a planar kinematic model. This step reduces the problem to a two-dimensional workspace, where the end-effector is required to reach an arbitrary target position (x, y) within a plane. The solution to this planar problem forms the foundation for the subsequent three-dimensional extension.

a) Kinematic Model: As illustrated in Fig. 8, the robotic arm is modeled as a two-link serial manipulator with link lengths L_1 and L_2. The base of the manipulator is fixed at the origin, and the end-effector position is defined as point $A(x, y)$.

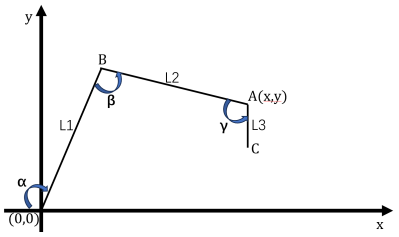


Fig. 8. Planar two-link manipulator model showing link lengths and joint angles.

The joint angles are defined using the parameters α and β, which are related to the actual joint angles θ_1 and θ_2 through the following transformations:

$$\theta_1 = \pi - \alpha \tag{1}$$

$$\theta_2 = \beta - \alpha \tag{2}$$

These definitions are chosen to align the mathematical formulation with the physical configuration of the servo motors.

b) Forward Kinematics: The forward kinematics describes the mapping from joint space (α, β) to Cartesian space (x, y). Based on the geometry of the manipulator, the end-effector position can be expressed as:

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_2) \tag{3}$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_2) \tag{4}$$

Substituting the angle definitions, we obtain:

$$x = L_1 \cos(\pi - \alpha) + L_2 \cos(\beta - \alpha) \tag{5}$$

$$y = L_1 \sin(\pi - \alpha) + L_2 \sin(\beta - \alpha) \tag{6}$$

Using trigonometric identities:

$$\cos(\pi - \alpha) = -\cos\alpha, \quad \sin(\pi - \alpha) = \sin\alpha \tag{7}$$

the forward kinematics can be rewritten as:

$$x = -L_1 \cos\alpha + L_2 \cos(\beta - \alpha) \tag{8}$$

$$y = L_1 \sin\alpha + L_2 \sin(\beta - \alpha) \tag{9}$$

These equations fully describe how the end-effector position depends on the joint angles.

c) Inverse Kinematics: To control the robotic arm, we must solve the inverse kinematics problem: given a target position (x, y), determine the corresponding joint angles α and β.

We first compute the radial distance from the origin:

$$r = \sqrt{x^2 + y^2} \tag{10}$$

Using the law of cosines, the angle β can be obtained as:

$$\beta = \arccos\left(\frac{L_1^2 + L_2^2 - r^2}{2L_1 L_2}\right) \tag{11}$$

Next, we compute the orientation angle of the target point:

$$\phi = \text{atan2}(y, x) \tag{12}$$

We then define an auxiliary angle γ:

$$\gamma = \arccos\left(\frac{L_1^2 + r^2 - L_2^2}{2L_1 r}\right) \tag{13}$$

Finally, the joint angle α is obtained as:

$$\alpha = \pi - (\phi + \gamma) \tag{14}$$

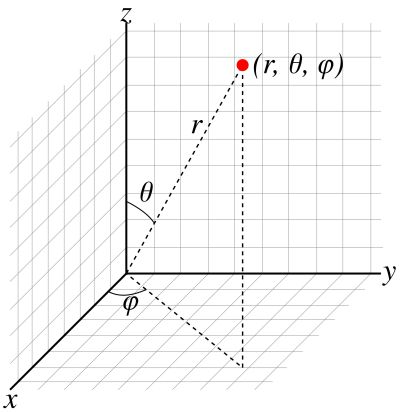


Fig. 9. Transformation from Cartesian coordinates to cylindrical coordinates and decomposition into planar motion and base rotation ϕ and γ.

d) Angle Continuity and Solution Selection: It is important to note that the inverse kinematics problem admits multiple solutions (e.g., elbow-up and elbow-down configurations). In practice, abrupt switching between these solutions can cause discontinuities in motion.

To ensure smooth and stable control, a continuity constraint is introduced by selecting the solution that minimizes the difference from the previous joint configuration. This approach avoids sudden jumps in joint angles and results in a continuous motion trajectory.

e) Summary: Through the combination of forward and inverse kinematics, the planar control problem is fully solved. The resulting formulation allows the system to directly map a desired Cartesian position (x, y) to the corresponding joint angles (α, β), enabling precise positioning of the end-effector within the plane. This planar model serves as the basis for the extension to three-dimensional motion described in the following section.

C. 3D Coordinate Transformation

After establishing the planar kinematic model, the next step is to extend the robotic arm to operate in three-dimensional space. While the planar model allows the end-effector to reach any target position within a two-dimensional workspace, real-world locomotion requires full spatial maneuverability.

a) Motivation for Coordinate Transformation: The key challenge lies in bridging the gap between the desired control input and the mechanical structure of the robot. From a control perspective, the robot is commanded using Cartesian coordinates (x, y, z), which represent a target position in three-dimensional space. However, the robotic arm itself is not directly actuated in Cartesian space. Instead, its motion is governed by:

- A base rotation around the vertical axis
- A planar two-link mechanism operating in a vertical plane

This mismatch requires a coordinate transformation that maps the Cartesian target into a representation that aligns with the robot's physical degrees of freedom.

b) Cylindrical Coordinate Representation: To resolve this mismatch, we introduce a cylindrical coordinate system. In cylindrical coordinates, a point in space is represented as:

$$(\rho, \theta, Z) \tag{15}$$

where:

- ρ is the radial distance from the origin in the horizontal plane
- θ is the rotation angle about the Z-axis
- Z is the vertical position

The transformation from Cartesian coordinates (x, y, z) to cylindrical coordinates is given by:

$$\rho = \sqrt{x^2 + y^2} \tag{16}$$

$$\theta = \text{atan2}(y, x) \tag{17}$$

$$Z = z \tag{18}$$

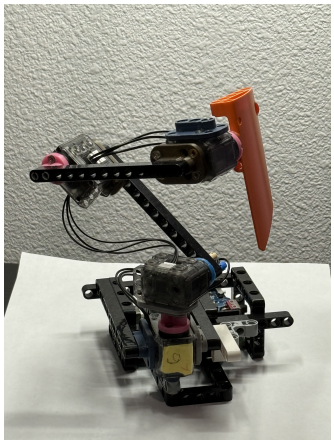


Fig. 10. Mechanical Arm

c) Mapping to Actuator Space: Once the cylindrical representation is obtained, the control problem can be decomposed into two independent components that correspond directly to the robot's actuators.

1) Base Rotation

The angular coordinate θ represents the rotation of the arm around the vertical axis. This value can be directly assigned to the base servo:

$$\theta_{\text{base}} = \theta \tag{19}$$

This allows the robot to orient itself toward the target position in the horizontal plane.

2) Planar Motion

The remaining coordinates (ρ, Z) define a point in a vertical plane. This reduces the three-dimensional problem back to the two-dimensional kinematic problem solved in the previous section.

Thus, the planar inverse kinematics is applied:

$$(\rho, Z) \rightarrow (\alpha, \beta) \tag{20}$$

where α and β are computed using the inverse kinematics formulation derived earlier.

d) Overall Control Pipeline: Combining all transformations, the full control pipeline can be expressed as:

$$(x, y, z) \rightarrow (\rho, \theta, Z) \rightarrow (\theta_{\text{base}}, \alpha, \beta) \rightarrow \text{servo commands} \tag{21}$$

This formulation provides a direct mapping from a desired three-dimensional target position to the joint angles required to achieve that position.

e) System Interpretation: From a geometric perspective, this transformation can be interpreted as rotating the planar workspace around the Z-axis. The original 2D arm operates within a plane defined by (ρ, Z), while the additional rotational degree of freedom θ allows this plane to sweep across the entire three-dimensional space.

This design offers several advantages:

- It reuses the planar inverse kinematics without modification
- It simplifies the 3D problem into independent subproblems
- It aligns naturally with the mechanical structure of the robot

As a result, the robotic arm is able to reach any point within its reachable workspace by combining base rotation with planar motion, while maintaining computational efficiency and implementation simplicity.

D. Communication and Control

After establishing the kinematic model and coordinate transformation pipeline, the robotic arm control can be expressed as a high-level function that maps a desired Cartesian target position (x, y, z) to the corresponding joint angles. However, to enable real-time interaction and flexible control, it is necessary to establish a reliable communication mechanism between the high-level controller and the embedded system.

a) System Architecture: The system adopts a master–slave architecture, where:

- The **high-level controller** (MATLAB or Python) is responsible for:
  - generating target positions (x, y, z)
  - computing auxiliary control parameters (e.g., orientation offset)
  - formatting and transmitting control commands
- The **low-level controller** (Arduino) is responsible for:
  - receiving and parsing incoming data
  - executing coordinate transformation and inverse kinematics
  - generating servo commands for actuation

Communication between the two layers is implemented using a **UART-based serial interface**, enabled through a USB-to-serial connection.

b) Communication Protocol Design: To ensure robust and unambiguous data transmission, a frame-based communication protocol is defined. Each command is transmitted as a structured data packet:

$$\texttt{\#X,Y,Z,error\$} \tag{22}$$

where:

- `#` denotes the start of a frame
- `$` denotes the end of a frame
- X, Y, Z represent the target Cartesian coordinates
- $error$ represents an auxiliary control parameter for end-effector orientation

A typical transmitted message is:

$$\texttt{\#+05.0,-06.0,+07.0,+30.0\$} \tag{23}$$

This explicit framing mechanism allows the receiver to reliably detect the beginning and end of each command, even in the presence of transmission noise.

c) Fixed-Width Data Encoding: To further improve parsing reliability, each floating-point value is encoded using a fixed-width format:

$$\texttt{\%+06.1f} \tag{24}$$

This encoding enforces:

- a constant field width
- an explicit sign ($+$ or $-$)
- a fixed decimal precision

As a result, the transmitted data maintains a consistent structure, reducing ambiguity during parsing and minimizing the likelihood of misinterpretation due to timing or buffering issues.

d) Serial Communication Parameters: The UART communication is configured with the following parameters:

- Baud rate: 38400 bps
- Data format: ASCII text stream
- Frame structure: start marker `#`, end marker `$`
- Line termination: newline characters (`\r`, `\n`) are ignored

The selected baud rate provides a balance between transmission speed and stability, enabling real-time updates at approximately 80–90 Hz while avoiding buffer overflow or synchronization errors.

e) Frame Parsing and Data Extraction: On the Arduino side, incoming serial data is processed using a state-based parser. The parsing procedure follows these steps:

1) Wait for the start marker `#`
2) Accumulate incoming characters into a buffer
3) Detect the end marker `$`
4) Split the buffer using delimiter `,`
5) Convert each token into floating-point values using standard functions

If any step fails (e.g., missing tokens or buffer overflow), the frame is discarded to prevent invalid commands from affecting system behavior.

f) Integration with Motion Control: Once a valid frame is received, the parsed values $(X, Y, Z, error)$ are directly passed into the control function:

$$(X, Y, Z, error) \rightarrow \texttt{moveToXYZ}(X, Y, Z, error) \tag{25}$$

This function performs:

- Cartesian to cylindrical coordinate transformation
- inverse kinematics computation
- joint angle generation
- servo command execution

Thus, the communication system establishes a complete control pipeline from high-level commands to physical actuation.

g) System Significance: The communication framework provides several important advantages:

- decouples high-level planning from low-level execution
- enables real-time interactive control
- ensures robust and consistent data transmission
- supports integration with multiple input sources (MATLAB, Python, vision systems)

From a system perspective, this design transforms the robotic arm into a flexible and extensible platform that can be driven by arbitrary high-level inputs, enabling both manual control and automated motion generation.

E. System Extension

Building upon the kinematic modeling and communication framework, the robotic arm control system is further extended to support interactive control, visualization, trajectory recording, and vision-based operation. These extensions significantly enhance the usability and functionality of the system, transforming it from a basic actuator-driven mechanism into a versatile platform for motion exploration and human–robot interaction.

a) High-Level Control Abstraction: At the system level, the entire motion control pipeline is encapsulated into a high-level function. From the perspective of the upper-level controller (MATLAB or Python), the user only needs to provide a target position in Cartesian space:

$$(x, y, z) \tag{26}$$

This target is transmitted to the Arduino controller via the communication protocol described in the previous section. The Arduino then performs all necessary coordinate transformations and inverse kinematics computations, and directly drives the servos to move the end-effector to the desired position.

This abstraction decouples user interaction from low-level control, greatly simplifying system usage.

b) Workspace Visualization in MATLAB: To provide intuitive feedback on the arm's motion, a three-dimensional visualization interface is implemented in MATLAB. Since the robotic arm consists of two links with lengths $L_1 = 10$ and $L_2 = 10$, the maximum reachable workspace is a sphere of radius 20 centered at the origin.

Accordingly, a spherical workspace is rendered in MATLAB to represent the feasible region of motion. The current target position of the end-effector is displayed as a red point within this sphere. This allows the user to directly observe the spatial position of the commanded target and understand how it relates to the robot's reachable workspace.

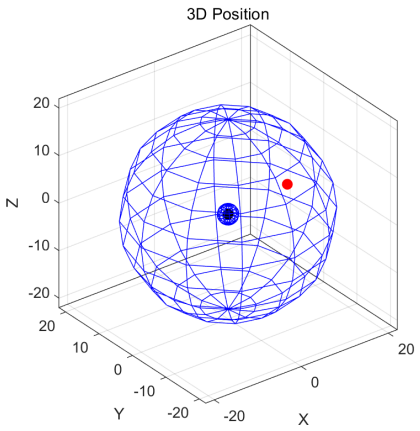


Fig. 11. MATLAB visualization of the reachable workspace and current target position of the robotic arm.

c) Interactive Control Interface: In addition to visualization, an interactive control panel is implemented in MATLAB to enable real-time manual control of the robotic arm. The interface maps user input directly to spatial motion:

- Mouse movement in the horizontal plane controls the x and y coordinates
- Mouse scroll wheel controls the z coordinate

This design allows the user to intuitively manipulate the end-effector in three-dimensional space using standard input devices. Compared to manually specifying coordinates, this approach significantly improves control efficiency and usability.

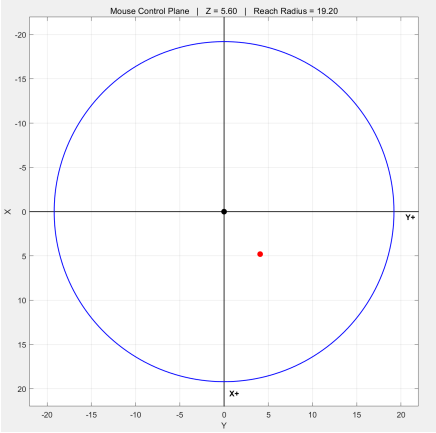


Fig. 12. MATLAB control interface for real-time manipulation of the robotic arm using mouse input.

d) Trajectory Recording and Gait Exploration: An additional feature of the MATLAB interface is the ability to record the trajectory of the end-effector during manual operation. As the user controls the arm, the system logs the sequence of positions over time, forming a trajectory in three-dimensional space.

This capability enables:

- replaying previously executed motions
- analyzing movement patterns
- identifying efficient locomotion strategies

In particular, this feature is valuable for studying robot gait design. By recording and comparing different motion trajectories, the user can iteratively refine movement strategies and optimize the robot's climbing performance.

e) Vision-Based Control Extension: To further enhance the system, a vision-based control interface is developed using real-time hand tracking. A computer vision library is used to detect hand landmarks from a live camera feed, extracting key joint positions in real time.

These detected hand positions are mapped to spatial coordinates, which are then transmitted to the Arduino using the same communication protocol as the MATLAB interface. As a result, the robotic arm can be controlled directly through hand gestures, replacing traditional input devices such as the mouse.

This extension introduces a more natural and intuitive form of interaction, enabling the user to control the robot through direct physical gestures.

f) System Significance: These system extensions significantly improve the flexibility and usability of the robotic platform. By integrating visualization, interactive control, trajectory recording, and vision-based input, the system evolves from a simple position-controlled manipulator into a comprehensive experimental platform.

This enables not only real-time control, but also systematic exploration of motion strategies, making it particularly suitable for studying locomotion behaviors and optimizing robotic performance in complex environments.

IV. Results and Discussion

Based on the experimental results presented in the previous sections, we summarize the overall system performance and analyze the key limitations and advantages of the proposed locomotion strategies.

a) Failure of the Robotic Arm Integration: Although a robotic arm system was designed to enhance the flexibility of locomotion, it was ultimately not integrated into the final climbing robot. In practice, the robotic arm exhibited two major limitations:

- Insufficient positioning accuracy, which led to unreliable engagement with surface features
- Limited force output, which prevented the arm from effectively pulling the robot body under load

Due to these constraints, the robotic arm was unable to provide stable and reliable locomotion in the tested environment. As a result, the final implementation relied primarily on the hook-based locomotion system.

b) Limitations of the Hook-Based System: While the hook-based design successfully enabled climbing on structured surfaces, several critical limitations were observed during experiments.

1) Unreliable Claw Engagement

The first major issue is that the claws often fail to achieve stable geometric interlocking with the ridges. Although multiple claw geometries were tested, the robot frequently experienced incomplete or unstable engagement. This results in slipping events and reduced climbing efficiency.

2) Lack of Directional Control

The second limitation is the lack of control over the robot's motion direction. The ridge structures on the surface are not strictly aligned, but instead form a fan-like distribution. As a result, small deviations in claw engagement can cause the robot to gradually change its heading.

Once the robot deviates from the intended climbing direction, it becomes increasingly difficult to correct its trajectory due to the passive nature of the design.

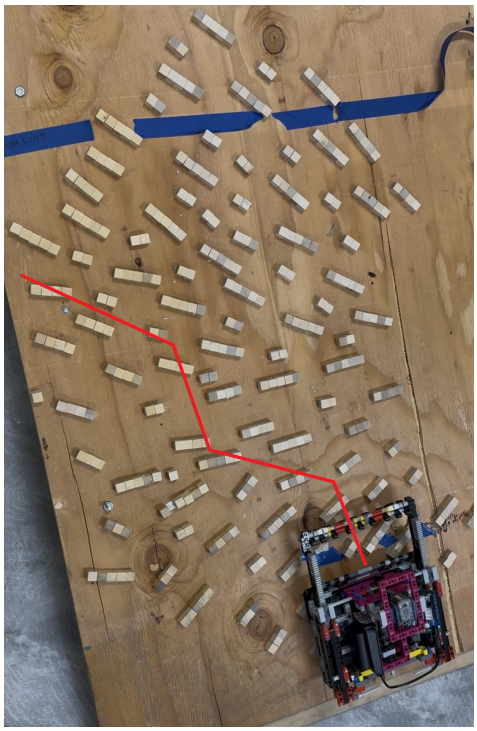


Fig. 13. Example trajectory of the robot during climbing, showing deviation from the intended direction over time.

As shown in Fig. 13, the robot gradually drifts away from the desired path, illustrating the difficulty of maintaining directional stability.

3) Low Locomotion Speed

Another observed limitation is the relatively slow climbing speed. The robot relies on a cyclic extension–engagement–retraction process, which inherently limits its speed. Additionally, frequent slips further increase the total traversal time.

c) Advantages of the System: Despite these limitations, the system demonstrates several important strengths.

1) Capability on Steep Slopes

The most notable advantage of the robot is its ability to operate on relatively steep inclines. Compared to other locomotion strategies, the hook-based mechanism remains functional even at slope angles up to $45°$, where many alternative designs fail.

2) High Force Output

The use of a servo-driven gear reduction mechanism allows the robot to generate sufficient force to overcome gravitational loading. Although this reduces speed, it ensures that the robot can maintain upward motion once proper engagement is achieved.

3) Robust Mechanical Simplicity

The passive nature of the design reduces system complexity and improves robustness. The robot does not rely on complex sensing or control algorithms, making it relatively stable under repeated operation.

d) Overall Discussion: The experimental results highlight a fundamental trade-off between simplicity, controllability, and performance. The hook-based system provides a mechanically simple and robust solution for climbing in structured environments, but suffers from limited controllability and efficiency.

At the same time, the robotic arm system, while theoretically more flexible, requires higher precision and force capabilities to be practically effective.

These findings suggest that future designs should aim to combine the robustness of passive mechanical engagement with the adaptability of active control, in order to achieve both stability and maneuverability in complex environments.

V. Future Work

While the proposed system demonstrates the feasibility of bio-inspired climbing on structured, low-friction surfaces, several limitations remain that suggest promising directions for future work.

a) Mechanical Design Improvements: One important direction is to improve the mechanical design of the claw system. The current design suffers from inconsistent engagement with surface ridges, which leads to slipping and reduced efficiency. Future work could explore optimized claw geometries, including variations in curvature, length, and multi-point contact configurations. In particular, extending the claw design from a one-dimensional arrangement to a two-dimensional grid structure may significantly improve stability and reduce the likelihood of directional drift.

Additionally, the robotic arm system could benefit from higher precision actuators and stronger torque output. Improving the accuracy and force capability of the arm would make it a more viable alternative to the passive hook-based system.

b) Control Strategy Enhancement: The experimental results indicate that the timing of actuation plays a critical role in locomotion performance. Future work should focus on developing more advanced control strategies, such as phase-synchronized control schemes, where auxiliary propulsion is automatically coordinated with the claw engagement cycle.

Furthermore, closed-loop feedback control could be introduced to dynamically adjust actuation based on real-time system states, such as slip detection or force estimation.

This would improve robustness and adaptability under varying environmental conditions.

c) Perception and Environment Awareness: Another important extension is to incorporate perception into the control loop. Currently, the robot operates without direct knowledge of the environment. By integrating vision-based sensing, the system could detect ridge locations and orientations, enabling more informed decisions about where and how to engage.

This would allow the robot to actively select optimal contact points, improving both efficiency and reliability.

d) Trajectory Optimization and Gait Learning: The trajectory recording capability developed in this project provides a foundation for future optimization. By analyzing recorded motion trajectories, it is possible to identify more efficient locomotion patterns.

Future work could explore automated gait optimization methods, including trajectory planning and learning-based approaches. For example, reinforcement learning or imitation learning could be used to discover optimal climbing strategies under different environmental conditions.

e) System Integration and Generalization: Finally, future work should aim to integrate the advantages of both locomotion strategies. A hybrid system that combines the robustness of passive claw engagement with the flexibility of an actively controlled robotic arm may provide a more general solution for complex terrains.

Such a system would be capable of adapting to a wider range of environments, balancing simplicity, stability, and controllability.

Overall, these directions highlight the potential for extending the current system into a more intelligent, adaptive, and high-performance robotic platform for locomotion in challenging environments.

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